

AI-Based Personal Finance Manager with Smart Expense Prediction System

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Title Justification

The title “**AI-Based Personal Finance Manager with Smart Expense Prediction System**” clearly describes the purpose, functionality, and uniqueness of the project.

- The term “**Personal Finance Manager**” indicates that the system is designed to help users manage their financial activities such as income tracking, expense monitoring, budgeting, and savings in an organized manner.
- The phrase “**AI-Based**” highlights the integration of Artificial Intelligence in the system. It emphasizes that the application goes beyond traditional finance management by incorporating machine learning techniques to provide intelligent insights.
- The component “**Smart Expense Prediction System**” represents the key innovation of the project. It signifies that the system can analyze past financial data and predict future expenses using algorithms like Random Forest, enabling users to make informed financial decisions.

Therefore, the title effectively reflects that the project is not just a basic finance tracker, but an **intelligent, data-driven system** that enhances financial planning through prediction and automation.

Abstract

- The AI-Based Personal Finance Manager with Smart Expense Prediction System is designed to help users manage their personal finances more effectively. Traditional expense tracking methods mainly record past transactions and provide limited support for financial planning. To overcome this limitation, the proposed system provides a web-based platform where users can track daily expenses, manage budgets, and monitor savings goals.
- The system analyzes historical spending data and predicts future expenses using the XGBoost Regression algorithm. It also provides expense analysis, spending insights, and budget alerts to help users understand their financial behavior and avoid overspending. The proposed system improves financial planning, supports better budgeting decisions, and promotes smarter money management through intelligent expense prediction.

Problem Statement

Traditional personal finance management methods mainly focus on recording past expenses and require users to manually track their financial activities. Existing expense tracking systems provide limited support for analyzing spending behavior and do not offer intelligent prediction of future expenses. Without proactive financial insights and automated budget monitoring, users often experience overspending, poor savings management, and difficulty in long-term financial planning. In addition, the absence of smart alerts and spending analysis makes it challenging for users to identify unusual financial patterns and maintain effective control over their personal finances.

Objectives of Project

OBJECTIVE 1

- To develop an AI-Based Personal Finance Manager that enables users to record, organize, and manage daily expenses, monitor category-wise budgets, and track savings goals through a centralized web-based platform for efficient financial management.

OBJECTIVE 2

- To design and implement machine learning models that analyze historical spending patterns, predict future expenses, generate budget alerts, and support users in improving financial planning and spending control.

Literature survey for objectives

No	Title	Author	Year	Methodology Adapted	Key Findings	Gaps
1.	AI-Powered Personal Finance Management: Predictive Analytics for Expense Forecasting	Sharma & Patel	2023	Machine learning models for expense prediction using financial data analysis	Predicts monthly expenses with high accuracy using historical spending patterns	Does not include full expense management and budgeting features
2	Intelligent Budget Optimization Using Machine Learning Algorithms	Chen, L., & Williams, R.	2022	Machine learning techniques for dynamic budget optimization	Improves budget planning by analyzing user spending behavior	Limited focus on user-friendly personal finance tracking
3	NLP-Based Automated Expense Categorization in Financial Applications	Rodriguez, M., & Kim, J.	2021	Natural Language Processing for automatic transaction categorization	Automatically categorizes expenses and improves financial data organization	Does not provide expense prediction

Proposed System

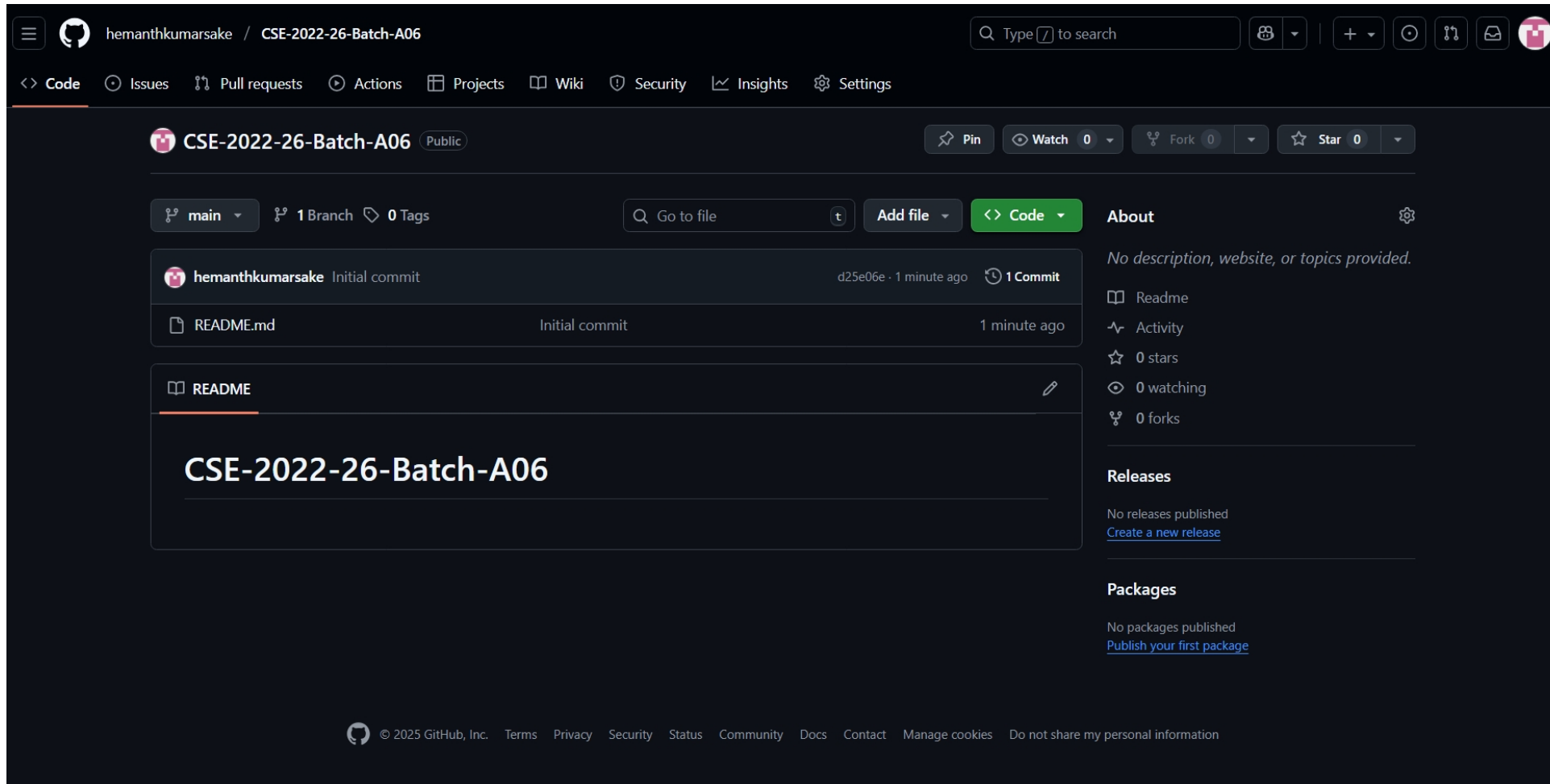
Our project presents an AI-Based Personal Finance Manager with Smart Expense Prediction System that integrates intelligent financial management features within a centralized web-based platform. The system consists of the following major modules:

- **Expense Management Module:** Allows users to record, update, and manage daily financial transactions by organizing expenses into categories such as food, travel, shopping, utilities, and entertainment for systematic financial tracking.
- **Budget and Savings Management Module:** Enables users to define monthly budgets and savings goals, continuously monitor spending behavior, and receive notifications when expenses approach predefined budget limits.
- **Smart Expense Prediction Module:** Uses the XGBoost Regression algorithm to analyze past spending patterns and predict future expenses for better financial planning.
- **Dashboard and Reporting Module:** Provides graphical reports, spending summaries, category-wise expense analysis, and prediction results through an interactive dashboard that helps users understand their financial behavior effectively.

References

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3. P. Gupta and M. Lee, “Behavioral Analytics in Personal Finance Management Systems,” *Journal of Behavioral Finance*, vol. 19, no. 4, pp. 301-317, 2022.
4. D. Wilson and X. Chen, “Django Framework for Scalable Financial Web Applications,” *Software: Practice and Experience*, vol. 52, no. 9, pp. 1876-1895, 2022.

Git Hub Dashboards of each student



Github Link : <https://github.com/hemanthkumarsake/CSE-2022-26-Batch-A06>

Any Queries?

Thank You!!!